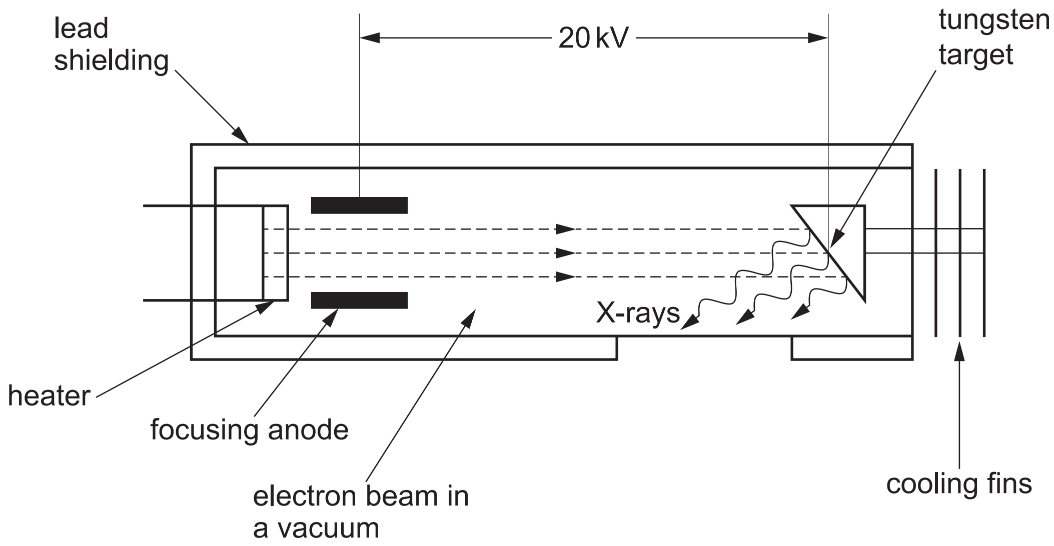
Examiner only

Option B – **Medical Physics**

8. (a) Below is a simplified diagram of the inside of an X-ray tube.



(i) State the main function of the heater **and** explain why there must be a vacuum between the heater and the tungsten target. [2]

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(ii) The electron beam has a current of 100 mA. Determine how many electrons arrive at the tungsten target per second. [2]

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(iii) Estimate the velocity of an electron in the X-ray tube as it hits the target. [3]

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(iv) Determine the minimum wavelength, λ_{min}, of the X-rays produced by this tube. [3]

(b) (i) An ultrasound probe can be used to study the speed of blood flow from the heart. Explain how the probe produces ultrasound **and** how the speed of blood flow can be determined. [3]

(ii) In measuring the speed of blood flow, the frequency of ultrasound used is 2.0 MHz and it travels through the blood at $1570\,\text{m}\,\text{s}^{-1}$. A frequency shift of 0.23 kHz is measured when the ultrasound is incident at an angle of 37° to the blood flow. Calculate the speed of the blood flow. [2]

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(c) Doctors are concerned that a patient is suffering from an overactive thyroid gland (hyperthyroidism). They have the choice of the following techniques to help diagnose this.

X-ray **MRI** **ultrasound** **CT scan** **radioactive tracers**

Evaluate the suitability of **all five** types of imaging techniques for detecting an overactive thyroid gland. [5]

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